

MEADOW LAKE, SK, Canada

WMO: 719700

Lat: 54.1286N

Lon: 108.5110W

Elev: 1578

StdP: 13.88

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.3	(c) -27.3	(d) -40.3	(e) 0.6	(f) -33.0	(g) -34.2	(h) 0.8	(i) -27.0	(j) 26.1	(k) 23.9	(l) 22.6	(m) 23.2	(n) 4.1	(o) 250	(p) 0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	82.9	64.9	79.7	63.6	76.7	62.2	68.0	78.9	66.0	76.2	64.1	73.5	10.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	95.7	72.9	62.0	88.2	70.5	60.2	82.5	68.6	33.3	78.9	31.6	76.3	30.2	73.6	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	19.4	17.4	-41.1	89.4	6.7	3.6	-45.9	92.0	-49.8	94.1	-53.5	96.1	-58.4	98.7
			-41.2	72.6	6.6	2.6	-46.0	74.5	-49.8	76.0	-53.5	77.5	-58.3	79.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.2	2.5	6.1	19.0	36.8	49.6	58.2	62.5	59.9	50.8	37.5	19.9	6.3
	DBStd	24.27	17.03	14.50	14.73	10.28	8.44	6.05	5.05	5.94	7.77	8.75	12.18	14.98
	HDD50	6929	1471	1228	962	407	113	8	0	4	81	397	902	1356
	HDD65	11289	1936	1648	1427	846	480	214	109	175	427	854	1352	1821
	CDD50	1179	0	0	0	11	100	255	387	312	106	8	0	0
	CDD65	64	0	0	0	0	2	11	31	18	2	0	0	0
	CDH74	1292	0	0	0	6	107	221	479	374	101	4	0	0
	CDH80	265	0	0	0	1	21	43	100	82	18	0	0	0
	WSAvg	7.9	7.5	7.4	8.0	8.7	8.9	8.2	7.4	7.1	8.0	8.2	7.7	7.4
	PrecAvg	18.4	0.8	0.6	0.7	1.1	1.4	3.5	3.3	2.7	1.7	1.2	0.8	0.7
(15) Precipitation	PrecMax	26.1	1.6	1.8	1.6	4.0	3.7	6.9	7.0	6.9	4.2	3.9	2.1	2.0
	PrecMin	12.9	0.2	0.1	0.1	0.0	0.1	1.5	1.0	0.7	0.2	0.2	0.0	0.2
	PrecStd	3.6	0.4	0.4	0.4	0.8	1.0	1.5	1.3	1.4	1.1	0.7	0.5	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	40.3	52.0	72.9	82.8	84.7	87.5	87.3	82.5	70.0	49.3	41.1
		MCWB	35.1	35.4	42.1	52.5	58.8	63.9	68.4	66.5	62.4	55.2	42.0	36.1
	2%	DB	36.1	35.3	44.4	65.5	76.6	79.6	82.5	81.7	76.7	61.6	43.4	36.1
		MCWB	33.2	31.7	37.6	49.0	56.2	61.9	66.8	65.3	60.2	49.4	38.3	32.8
	5%	DB	32.3	31.1	40.4	59.7	72.1	75.9	79.1	78.2	71.3	56.3	39.1	31.9
		MCWB	29.9	28.9	35.5	46.1	53.7	59.7	65.6	64.0	57.5	46.5	35.3	29.6
	10%	DB	27.0	27.5	37.3	54.4	67.8	72.5	76.0	74.6	66.4	51.4	35.4	27.7
		MCWB	25.4	25.6	33.5	43.5	51.9	58.6	63.9	62.3	54.9	44.1	32.6	26.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.7	35.7	43.1	53.8	61.0	67.5	72.0	70.0	64.7	55.7	42.7	36.4
		MCDB	38.6	39.6	51.4	70.0	76.1	79.0	81.7	81.3	78.0	69.0	48.2	40.0
	2%	WB	33.4	32.1	38.2	49.9	58.1	64.6	69.1	67.1	61.2	50.9	39.0	33.0
		MCDB	36.1	34.9	43.5	63.5	72.2	74.1	78.9	78.3	73.7	58.9	42.8	35.9
	5%	WB	29.9	28.9	35.8	46.8	55.7	62.4	66.9	65.1	58.5	47.5	35.5	29.6
		MCDB	32.3	31.0	39.9	58.3	68.8	72.3	76.9	75.7	69.4	55.0	38.7	31.9
	10%	WB	25.4	25.6	33.8	44.0	53.3	60.3	65.0	63.1	55.9	44.8	32.7	25.9
		MCDB	27.0	27.5	37.0	53.5	65.6	69.2	74.1	72.8	64.7	50.7	35.2	27.7
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	21.4	21.1	21.4	25.6	22.4	22.8	23.8	23.3	19.2	16.1	18.1
		MCDBR	20.7	22.1	21.6	31.0	33.4	29.5	28.9	31.2	33.3	28.7	20.5	21.0
		MCWBR	18.1	19.3	16.0	17.6	16.5	14.6	15.3	16.5	18.4	17.6	15.4	18.5
	5% WB	MCDBR	20.5	21.2	20.4	29.3	29.0	24.3	26.2	27.4	30.9	26.8	19.3	20.7
		MCWBR	18.0	18.8	15.4	17.4	15.2	13.4	15.0	15.6	17.8	17.2	15.0	18.4
(40) Clear-Sky Solar Irradiance	taub		0.242	0.244	0.292	0.320	0.345	0.365	0.369	0.361	0.311	0.286	0.250	0.217
	taud		2.328	2.371	2.353	2.431	2.403	2.367	2.377	2.383	2.525	2.560	2.469	2.365
	Ebn at Noon		237	279	284	290	286	280	276	272	276	261	237	229
	Edh at Noon		17	23	30	32	35	36	35	33	25	19	15	13
(44) All-Sky Solar Radiation	RadAvg		280	612	1096	1482	1805	1815	1862	1520	1064	605	272	197
	RadStd		27	42	75	121	158	133	121	122	121	101	37	19

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page